Structural calculation report

B. Piled raft, 4 x 4 at 2.4 m, 9 x 9 x 1.2 m, four columns of 2,500 kN D + 800 kN L

Item	
Model	B. Piled raft, 4 x 4 at 2.4 m, 9 x 9 x 1.2 m, four columns of 2,500 kN D + 800 kN L
Date	2026-10-24
Software	Pondus 0.1.3 with the Stabileo (dedaliano) engine
Engineer	Pondus example

1 Summary

The model has 20 nodes and 0 frame members (0 columns, 0 beams and 0 other members), 0 structural walls, 1 slabs, 16 supports, 2 load cases and 3 combinations. The design covers 0 beams, 0 columns, 0 wall storeys, 1 slabs, 0 footings, 0 steel members and 0 members of other materials, and the largest design ratio is 0.76 (slab Piled raft: punching at load at node 17). Every designed element passes its checks at the target ratio.

Reinforced concrete is designed to ACI 318-19 and structural steel to AISC 360-16 (LRFD). Timber members follow the NDS (ASD), masonry TMS 402 and cold-formed steel AISI S100 where the model has them.

2 Basis of design

Materials

#	Material	Kind	E (MPa)	γ (kN/m³)	f'c (MPa)	fy (MPa)	fu (MPa)
1	Concrete f'c 24 MPa	Concrete	23025	24.0	24.0	—	—
2	Steel SS400	Steel	200000	77.0	—	245	400

Design settings

Setting	Value
Main bars fy / stirrups fyt	390 / 235 MPa
Deflection limits (live / total)	L/360 / L/240
Target design ratio	1.00
Slab method	Plate
Second-order analysis (P-Δ) of ULS combinations	no

3 Model

The structure is a three-dimensional frame analysed by the direct stiffness method with the members split at every node, slab mesh point and load break, so the member forces below are exact for the frame idealisation. It has 1 levels at z = 0.00 m.

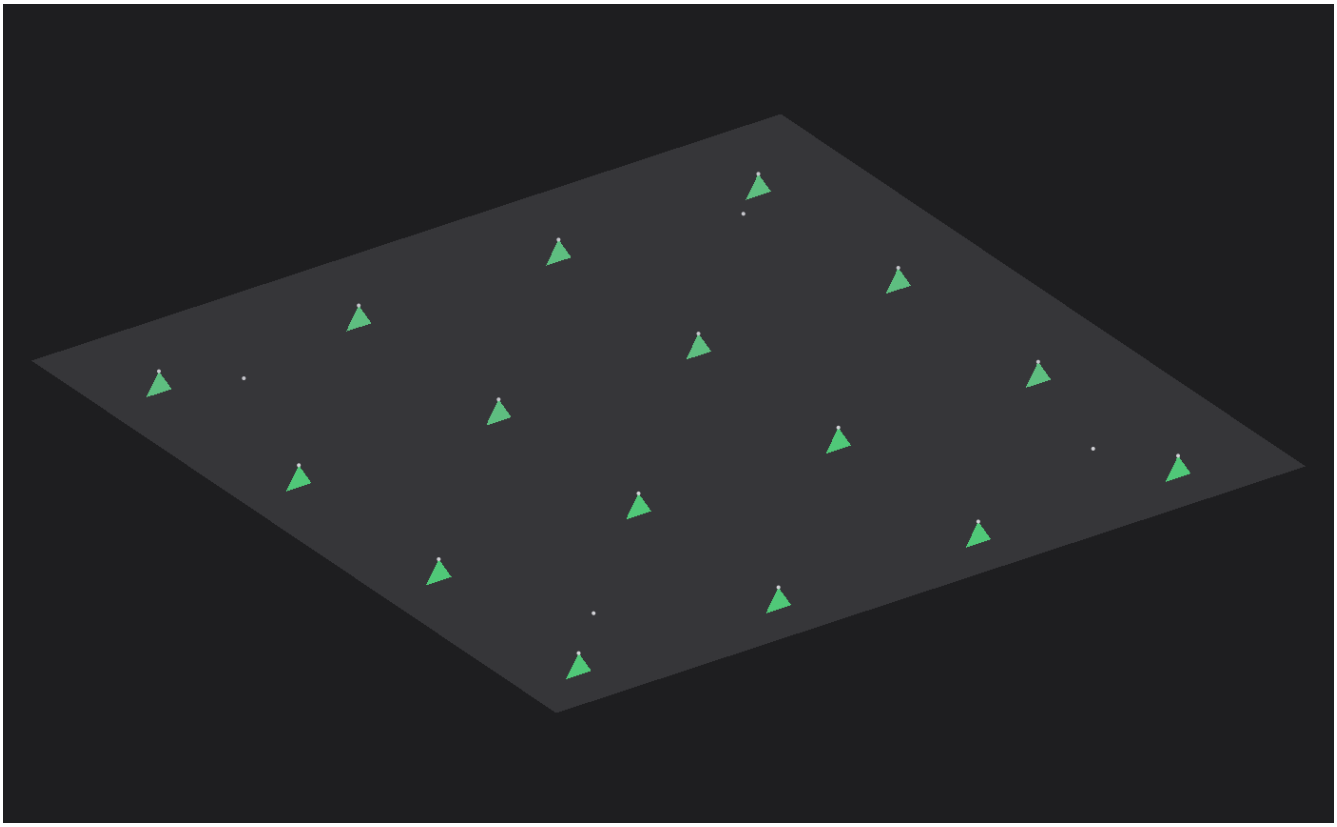


Figure 1. The model in 3D: columns, beams, braces and slabs.

Sections

Section	Members	A (m²)	Iy (m⁴)	Iz (m⁴)	J (m⁴)	Zy (m³)

Supports

The supports are 16 custom.

4 Loads and their balance

Each load case is solved alone, and the sum of the support reactions is compared with the applied load, slab loads and self-weight included, which checks that every load reached the supports.

Case	Kind	Self-weight	Applied Fz (kN)	Σ reactions Fz (kN)	Applied Fx	Applied Fy	Balanced
D	D	yes	-12332.80	12332.80	0.00	0.00	yes
L	L	no	-3200.00	3200.00	0.00	0.00	yes

Combinations

Combination	Kind	Factors
ULS 1.4D	Uls	1.4 D
ULS 1.2D + 1.6L	Uls	1.2 D + 1.6 L
SLS D + L	Sls	1 D + 1 L

5 Analysis results

The extremes of each combination are listed with the member where they occur. Moments are positive in sagging, axial forces positive in tension, and the displacement is the largest node movement.

Combination	My max (kN·m)	My min (kN·m)	|Vz| max (kN)	N min (kN)	δ max (mm)
ULS 1.4D	0.0 (m 0)	0.0 (m 0)	0.0 (m 0)	0.0 (m 0)	7.72
ULS 1.2D + 1.6L	0.0 (m 0)	0.0 (m 0)	0.0 (m 0)	0.0 (m 0)	8.94
SLS D + L	0.0 (m 0)	0.0 (m 0)	0.0 (m 0)	0.0 (m 0)	6.97

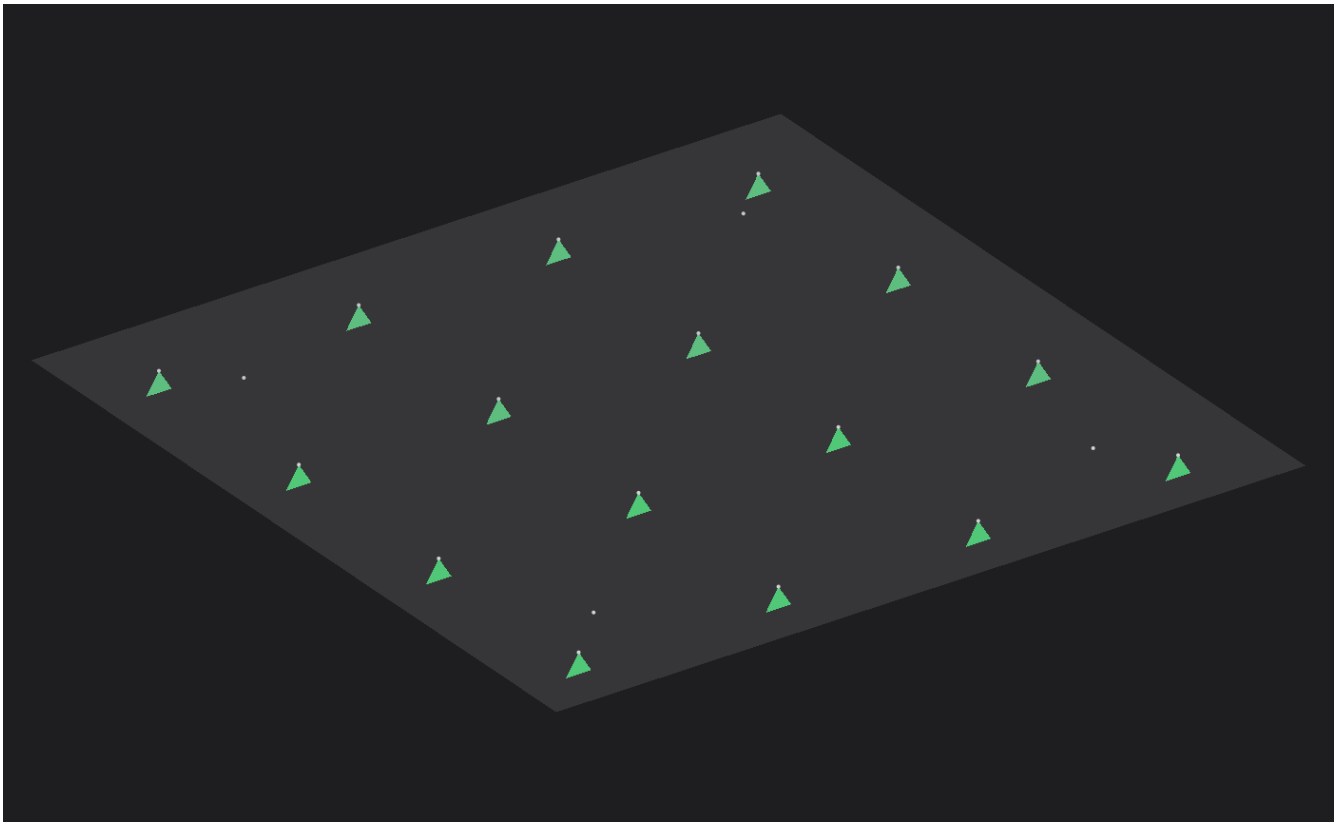


Figure 2. Bending moment My under 'ULS 1.4D', drawn on the tension side.

6 Design

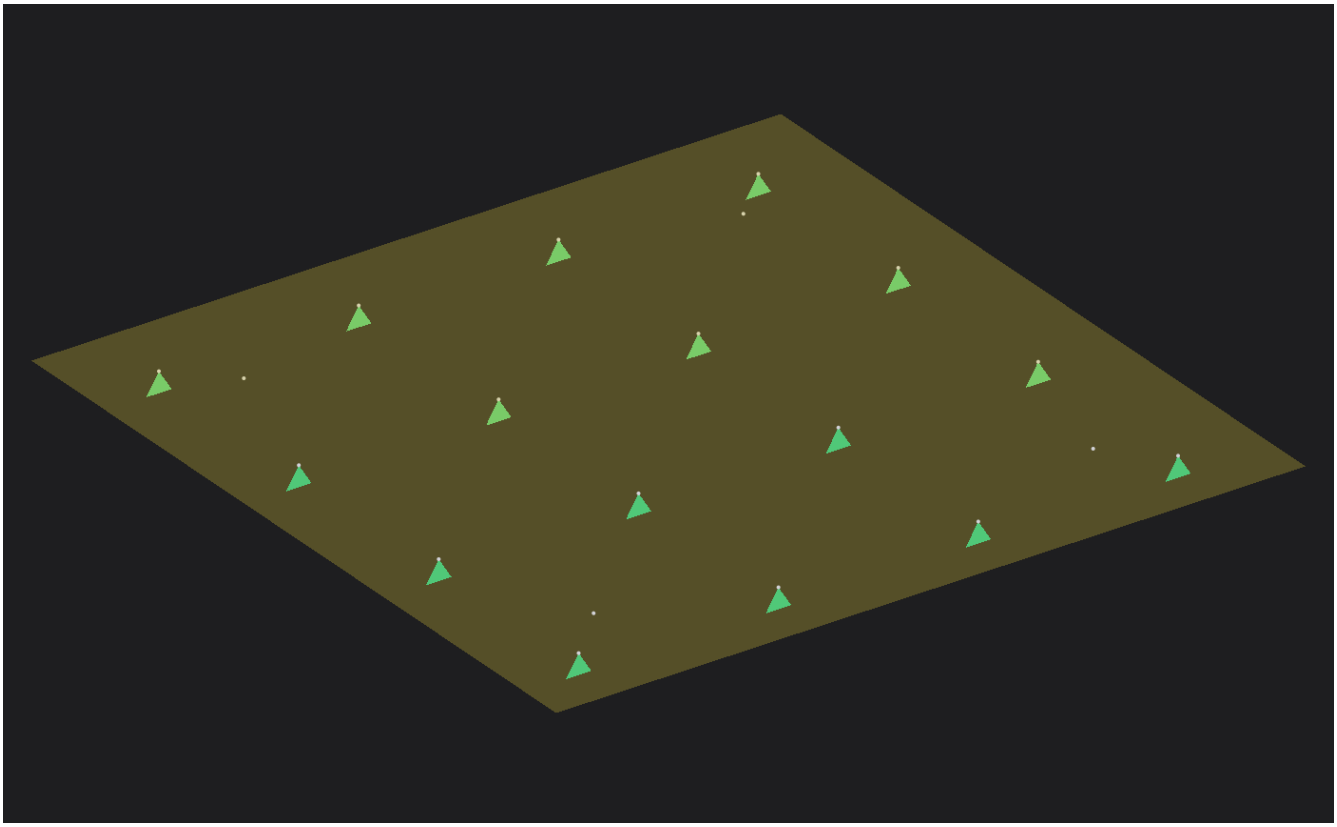


Figure 3. Design ratios: green below 0.7, yellow below 0.9, orange below 1.0 and red over 1.0.

Slabs

Slab	lx x ly (m)	h (mm)	Method	Bars	Ratio	Governs
Piled raft	9.00 x 9.00	1200	mat FE (MITC4, 441 elements, Wood–Armer, punching at 20 points)	x between supports (bottom): DB16 @ 75 mm; y between supports (bottom): DB16 @ 75 mm; x over supports (top): DB16 @ 75 mm; y over supports (top): DB16 @ 75 mm	0.76	punching at load at node 17

Calculations of the governing members

The tables below give the demand, the capacity and the ratio of each check of the most utilised member of each type, so the governing numbers can be followed by hand.

Slab Piled raft (9.00 x 9.00 m, h = 1200 mm, mat FE (MITC4, 441 elements, Wood–Armer, punching at 20 poin

Strip	Mu (kN·m/m)	As req (mm²/m)	As prov (mm²/m)	Bars	Ratio
x between supports (bottom)	805.27	2160	2681	DB16 @ 75 mm	0.74
y between supports (bottom)	805.27	2160	2681	DB16 @ 75 mm	0.75
x over supports (top)	640.65	2160	2681	DB16 @ 75 mm	0.59
y over supports (top)	640.65	2160	2681	DB16 @ 75 mm	0.60

The factored load is wu = 0.00 kPa, the shear Vu = 0.0 kN/m against φVc = 728.6 kN/m, and the minimum thickness is 0 mm.